Tutorial 10

(01) b

(02) c

(03) c

(04) c

(05) c

(06) c

(07) (a) $t_0 = 15$ $v(t_0) = 362.78$

$t_1 = 20$ $v(t_1) = 517.35$

$b_0 = v(t_0) = 362.78$

$$v(t) = v(t_0) + \frac{v(t_1) - v(t_0)}{t_1 - t_0}(t - t_0)$$

$$= 362.78 + \frac{(517.35 - 362.78)(16-15)}{20-15}$$

$$= 393.694$$

(b) $v(t) = a_1 t^2 + b_1 t + c_1$ $0 \le t \le 10$

$v(t) = a_2 t^2 + b_2 t + c_2$ $10 \le t \le 15$

$v(t) = a_3 t^2 + b_3 t + c_3$ $15 \le t \le 20$

$v(t) = a_4 t^2 + b_4 t + c_4$ $20 \le t \le 22.5$

$v(t) = a_5 t^2 + b_5 t + c_5$ $22.5 \le t \le 30$

$x = 0$

$c_1 = 0$ — (1)

$x = 10$

$100a_1 + 10b_1 + c_1 = 227.04$ — (2)

$100a_2 + 10b_2 + c_2 = 227.04$ (3)

$x = 15$

$225a_2 + 15b_2 + c_2 = 362.78$ — (4)

$225a_3 + 15b_3 + c_3 = 362.78$ — (5)

$x = 20$

$400a_3 + 20b_3 + c_3 = 517.35$ — (6)

$400a_4 + 20b_4 + c_4 = 517.35$ — (7)

$x = 22.5$

$506.25a_4 + 22.5a_4 + c_4 = 602.97$ — (8)

$506.25a_5 + 22.5a_5 + c_5 = 602.97$ — (9)

$x = 30$

$900a_5 + 30b_5 + c_5 = 901.67$ — (10)

$v'(t) = 2a_1 t + b_1$

$v'(t) = 2a_2 t + b_2$

$v'(t) = 2a_3 t + b_3$

$v'(t) = 2a_4 t + b_4$

$v'(t) = 2a_5 t + b_5$

We assume $a_1 = 0$

when (2),

$10b_1 = 227.04$

$b_1 = 22.704$

$20a_1 + b_1 = 20a_2 + b_2$ — (11)

$30a_2 + b_2 = 30a_3 + b_3$ — (12)

$40a_3 + b_3 = 40a_4 + b_4$ — (13)

$45a_4 + b_4 = 45a_5 + b_5$ — (14)

When (11),

$20a_2 + b_2 = 22.704$ — (15)

(4) − (3)

$125a_2 + 5b_2 = 135.74$ — (16)

(16) - (15) × 5

$25a_2 = 22.22$

$a_2 = 0.8888$

When (15),

$b_2 = 22.704 - 17.776$

$b_2 = 4.928$

When (12),

$30a_3 + b_3 = 31.502$ - (17)

(6) - (5)

$175a_3 + 5b_3 = 154.57$ - (18)

(18) - (17) × 5

$25a_3 = -3.39$

$a_3 = -0.1356$

When (17),

$b_3 = 35.66$

When (13)

$400a_4 + b_4 = 30.236$ - (19)

(8) - (7)

$106.25a_4 + 2.5b_4 = 85.62$ - (20)

(20) - (19) × 2.5

$6.25a_4 = 10.03$

$a_4 = 1.6048$

When (19),

$b_4 = -33.956$

When (14),

$45a_5 + b_5 = 38.26$ - (21)

(10) - (9)

$393.75a_5 + 7.5b_5 = 298.7$ - (22)

(22) - (21) × 7.5

$56.25a_5 = 11.75$

$a_5 = 0.2089$

When (21),

$b_5 = 28.86$

When (4),

$c_2 = 362.78 - 273.9$

$c_2 = 88.88$

When (5),

$c_3 = 362.78 + 30.51 - 534.7$

$c_3 = -141.97$

When (7),

$c_4 = 517.35 + 37.2$

$c_4 = 554.55$

When (9),

$c_5 = 602.97 - 755.11$

$c_5 = -152.14$